Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

Lecture 3 – Quantization, representation, and formats



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Begränsad delning

Repetition from Lecture 2

Repetition: Fourier transform expanded

- Continuous Fourier Transform (CFT)

$$X(f) = \mathcal{F}x(t) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt$$

- Inverse CFT

$$x(t) = \mathcal{F}^{-1}X(f) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\,df = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega$$

- Discrete-Time Fourier Transform (DTFT)

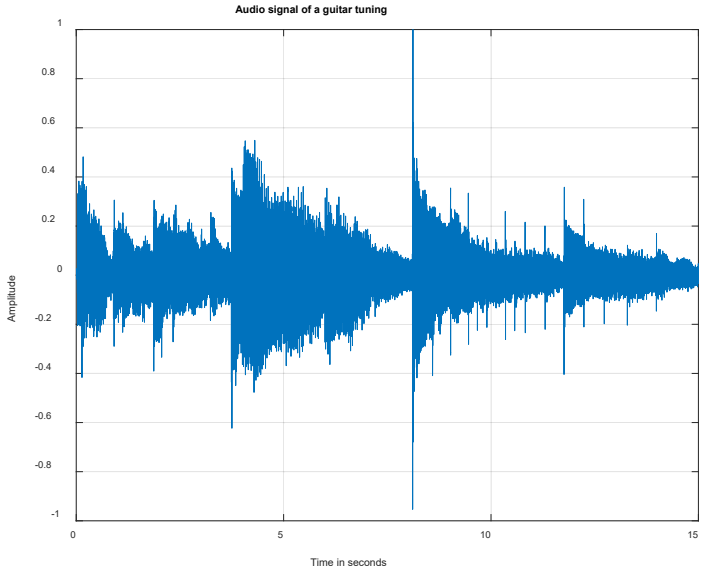
$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

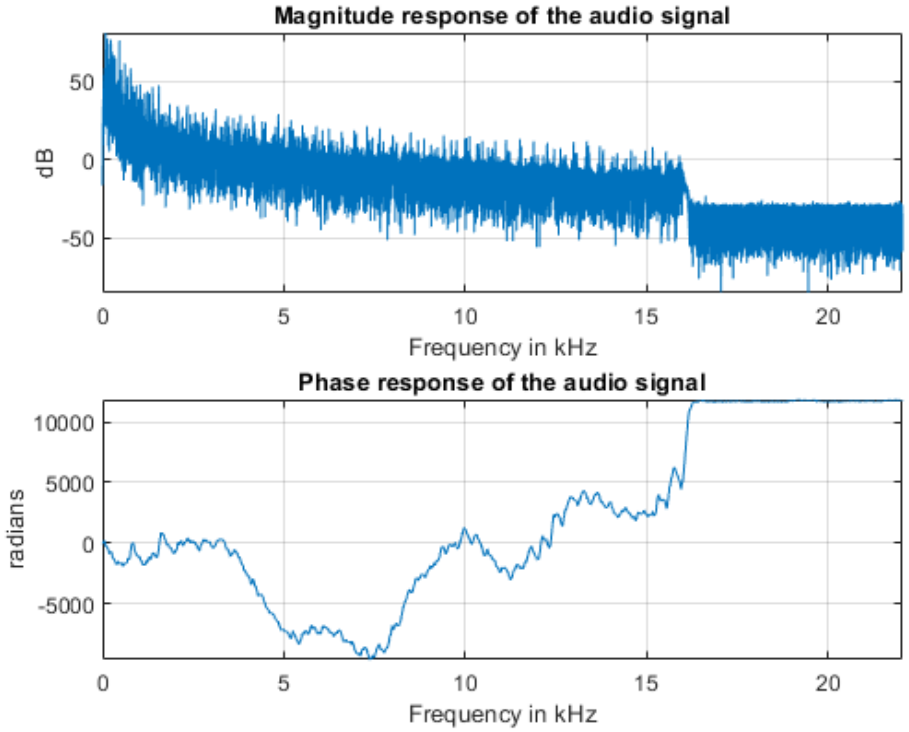
- Inverse DTFT

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n}\,d\Omega$$

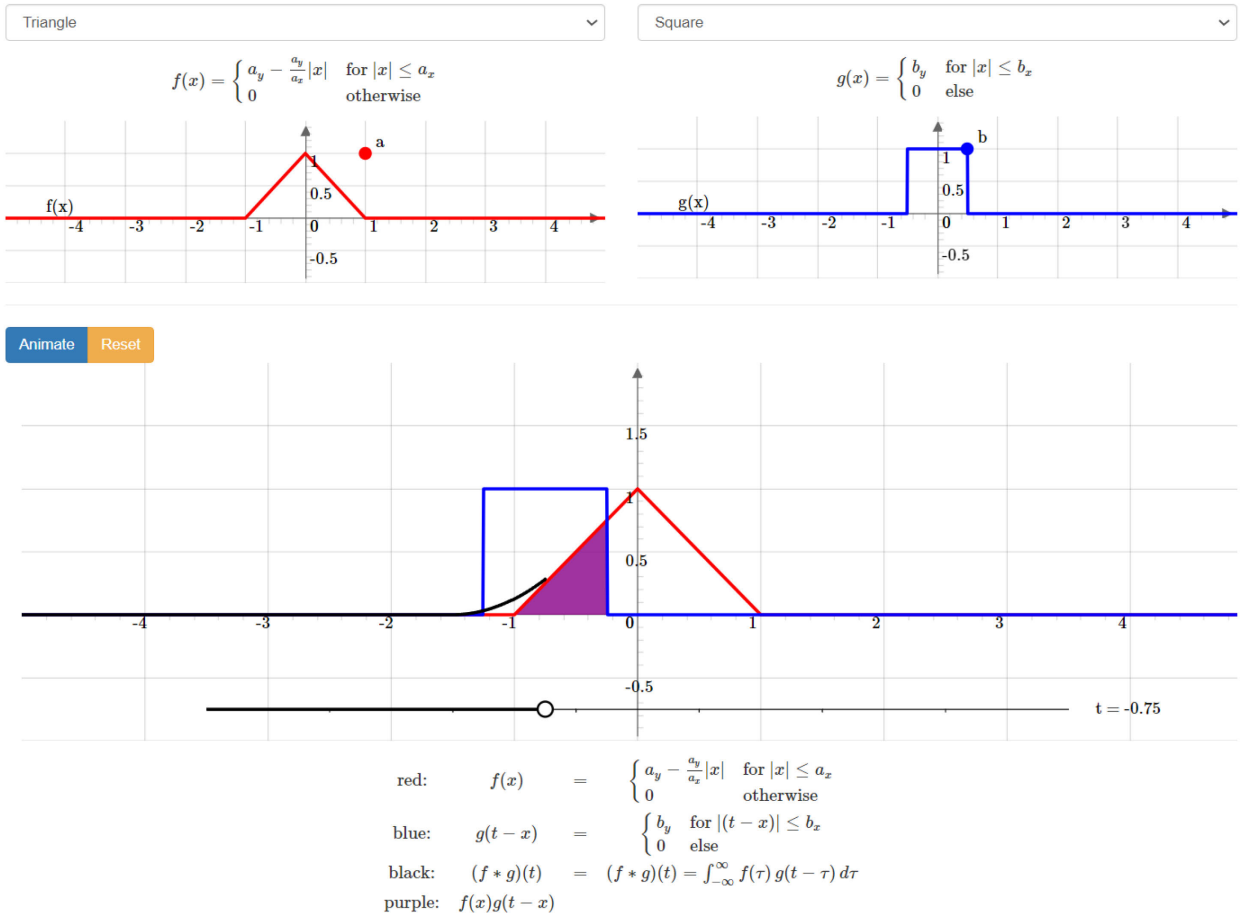
Repetition: Fourier transform – frequency domain representation

- Magnitude $|X(f)|$ Strength of each frequency component
- Phase $\angle X(f)$ Time delay of each frequency component





Repetition: Convolution



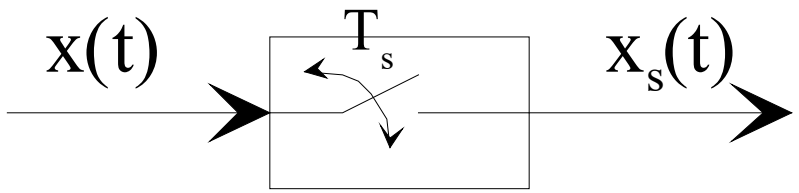
Demo

Source: phiresky's public GitHub repositories

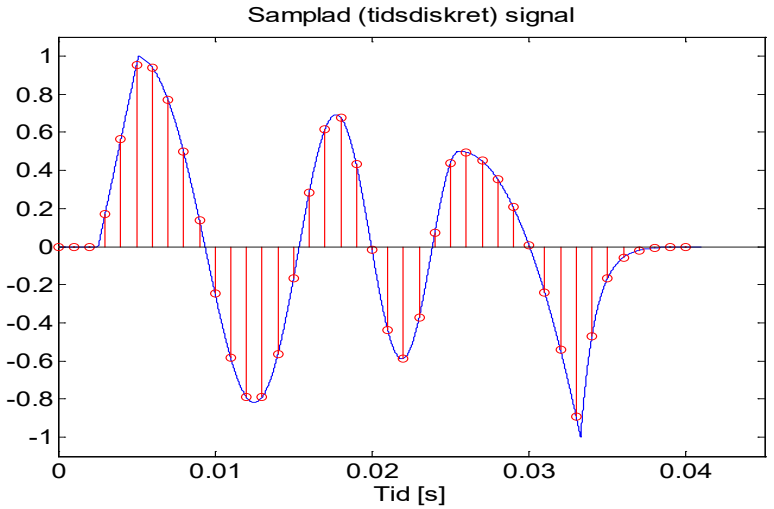
Repetition: Sampling

- Discretization in time
- Selects data for certain time instances.

$$x_s(n) = x(n \cdot T_s),$$
$$n = \ldots -1,0,1,2,3,\ldots$$



- T_s = sampling time,
- $fs = 1/Ts$ = sampling frequency
- [$\omega_s = 2\pi f_s$ = angular sampling frequency]

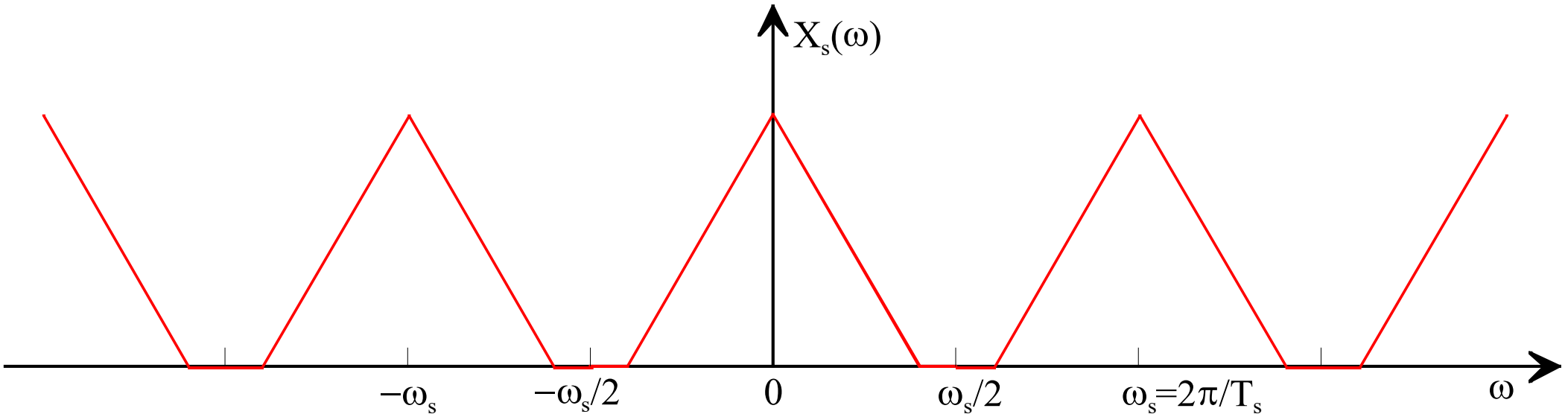


Repetition: Simpling implies Periodic spectrum

- A time-discrete (sampled) signal $x_s(n)$ always produces a periodic spectrum $X_s(\omega)$.
  - The original (continuous signal) spectrum $X_s(\omega)$ is repeated with periodicity $\omega_s = 2\pi/T_s$.

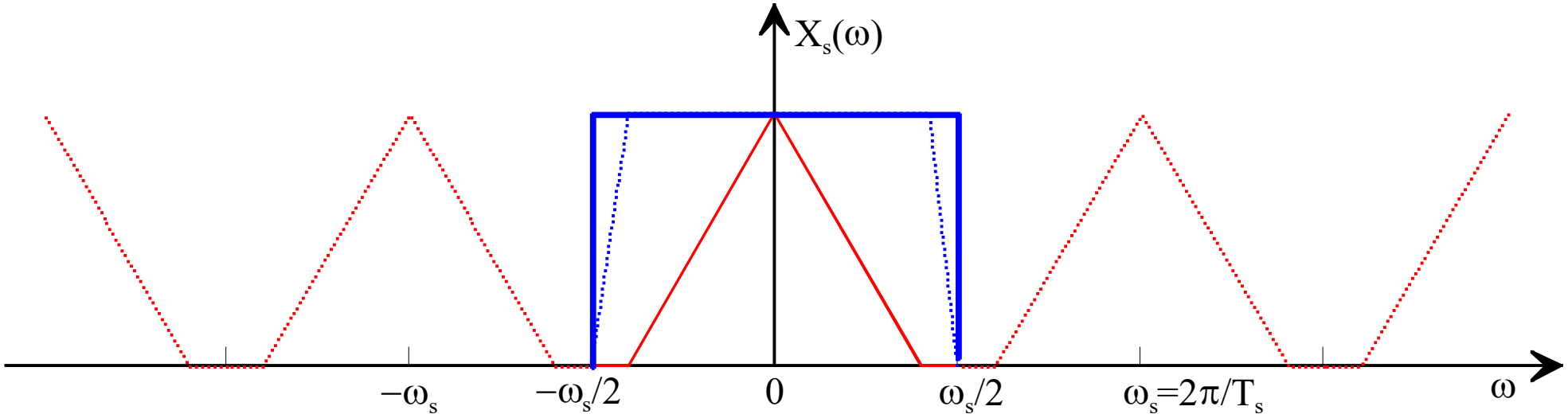
$$X_s(\omega) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} X(\omega - n\cdot\omega_s)$$

$$x_s(n) = x(n\cdot T_s),$$
$$n = \ldots -1,0,1,2,3,\ldots$$



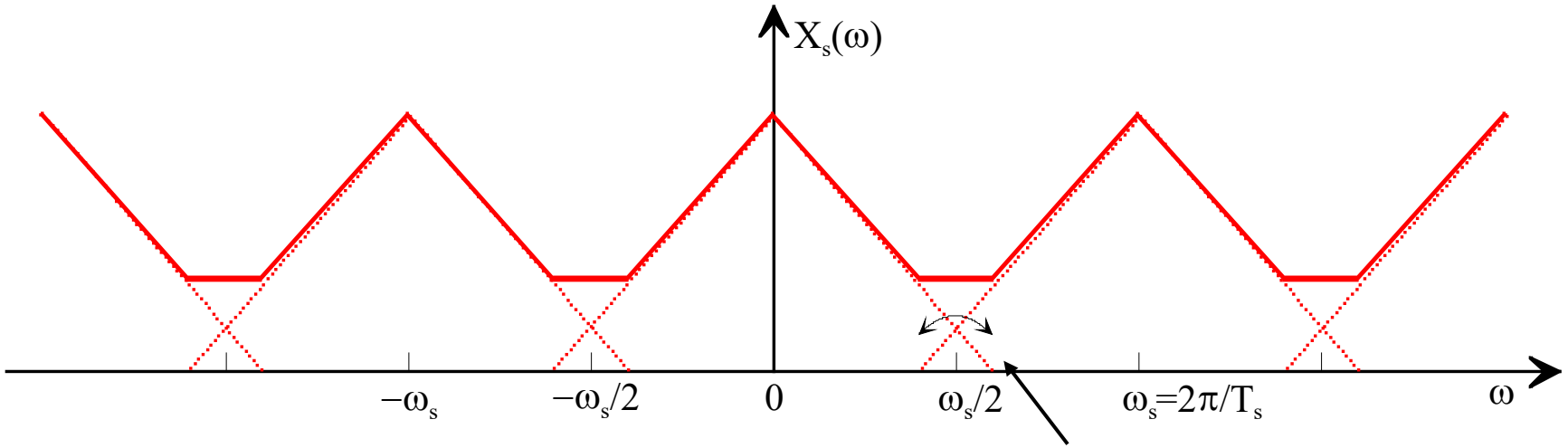
Repetition: Reconstruction

- Thereby, the original spectrum $X(\omega)$ is reconstructed
- Thus, the original signal $x(t)$is reconstructed



Repetition: Aliasing

- And what happens then with the spectrum $X_s(\omega)$?



The part of the spectrum $\omega > \omega_s/2$ is ”folded” and added

- Ideal LP => $X_{reconstructed}(\omega) \neq X(\omega)$!

Repetition: Nyquist-Shannon Sampling Theorem

- For a signal to be fully reconstructed from the sampled signal, the sampling frequency ω_s must be at least twice the largest frequency ω_{max} of the original signal.

$$\omega_s > 2\,\omega_{max}$$

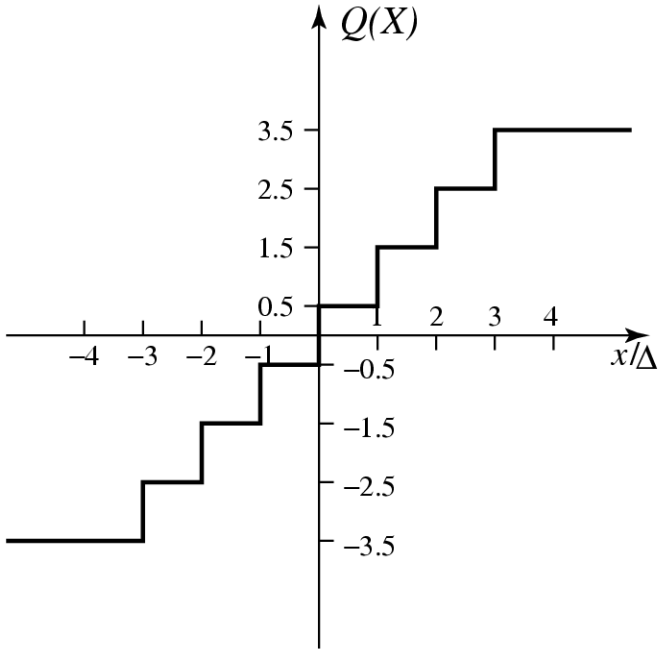
$$f_s > 2\,f_{max}$$

Begränsad delning

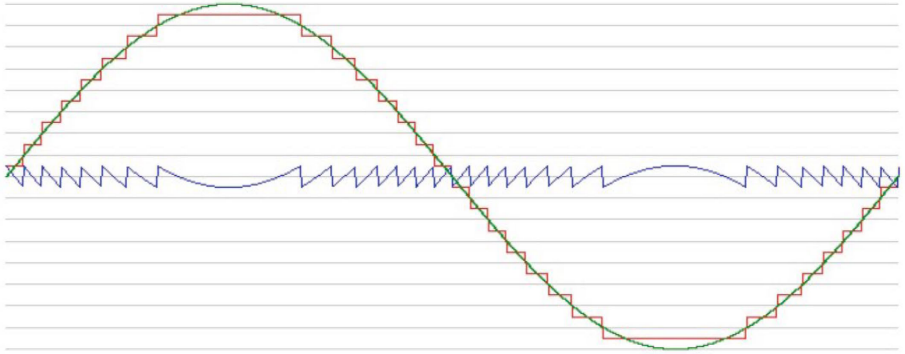
Quantization

Quantization

- Reducing the continuous sample values to a discrete number of values
- Introduces roundoff errors, so called quantization noise



- M – Number of quantization levels
- N – Number of bits
- $M = 2^N$



Continuous
Quantized
Quantization error

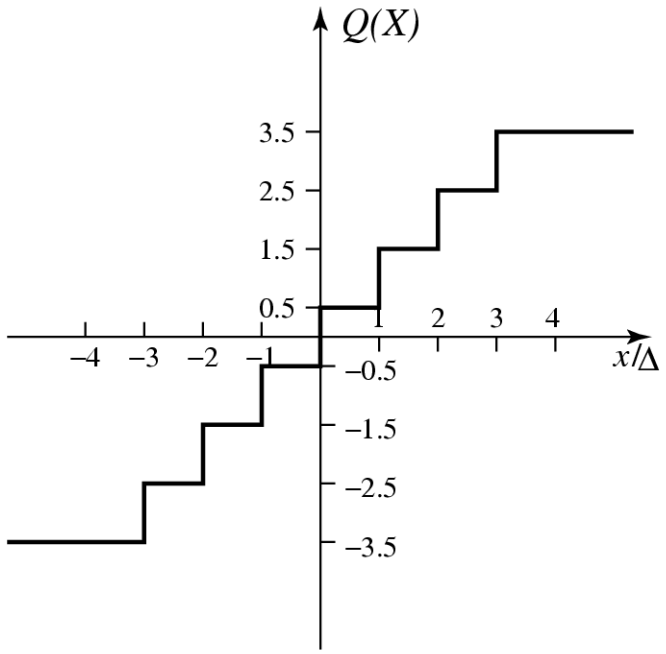
Demo

Quantization

Uniform scalar quantizer (assume step size $\Delta = 1$)

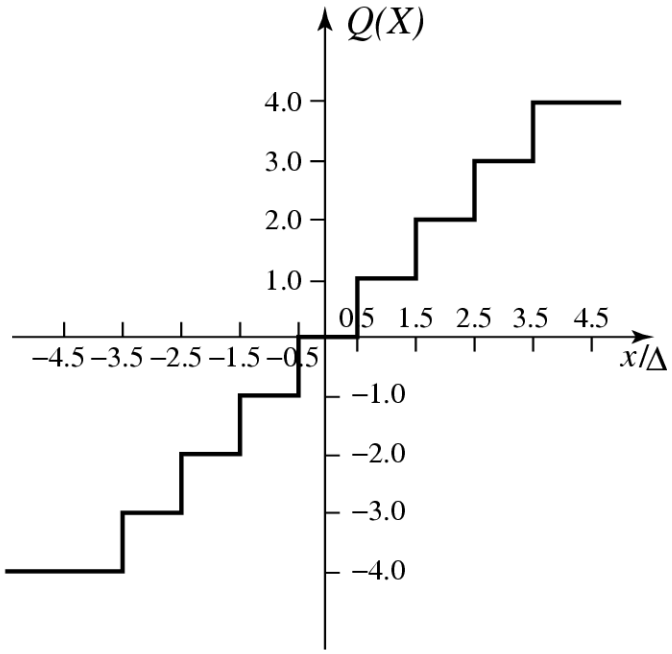
Midrise

$$Q_{midrise}(x) = \lceil x \rceil - 0.5$$



Midtread

$$Q_{midtread}(x) = \lfloor x + 0.5 \rfloor$$



Less noisy around x=0

Signal to Noise Ratio (SNR)

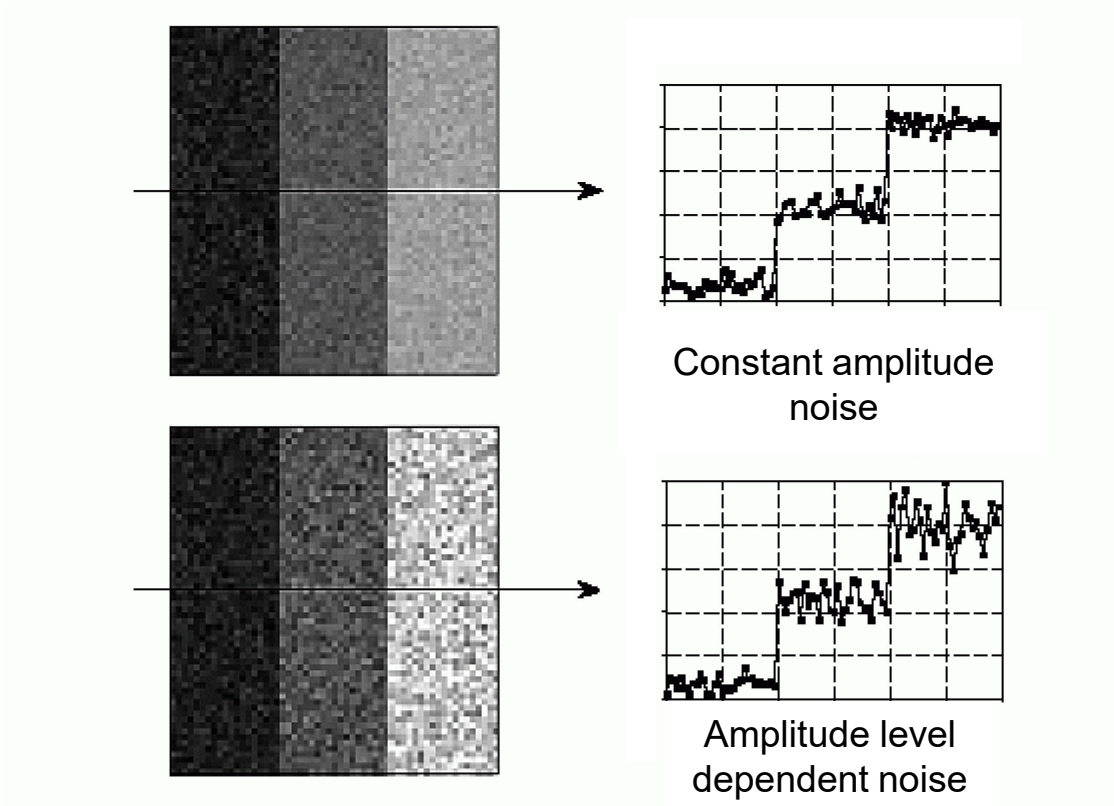
- A measure of the signal quality
  - Signal power related to noise power
  - Usually measured in decibel (dB)

$$SNR = 10\log_{10}\frac{V_{signal}^2}{V_{noise}^2} = 20\log_{10}\frac{V_{signal}}{V_{noise}}$$

- Signal power S is proportional to square of the signal amplitude V^2

$$S \propto V^2$$

  - E.g. if the amplitude is 10 times the noise, the SNR is $20 \cdot log_{10}(10) = 20dB$



Demo

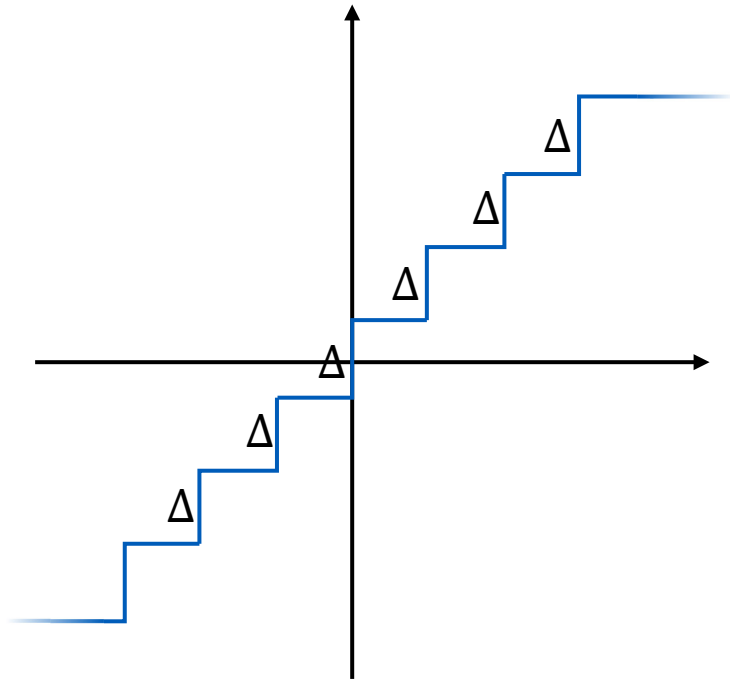
Source: [The Scientist and Engineer's Guide to Digital Signal Processing By Steven W. Smith, Ph.D.]

Signal to Quantization Noise Ratio (SQNR)

- Quantization noise V_{qn} is the difference between the actual analog value and the nearest quantization interval

- Distance between two quantization intervals is Δ

$$|V_{QN}| \leq \frac{\Delta}{2}$$

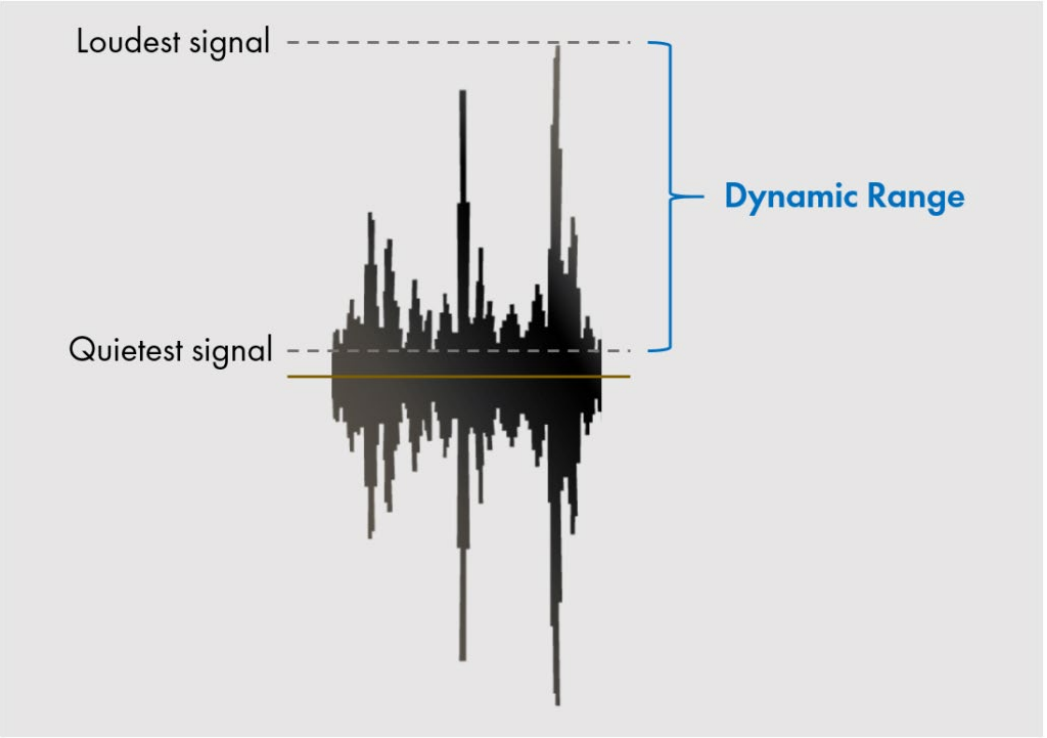


Demo

Dynamic range

- Signal dynamic range
  - Ratio of maximum to minimum *absolute values*

$$\frac{V_{max}}{V_{min}} = \frac{\max|V|}{\min|V|}$$



- For a signal quantized to N bits
  - V_{max} gets mapped to $2^{N-1} - 1$
  - V_{min} gets mapped to 1
  - $-V_{max}$ gets mapped to -2^{N-1}

- The quantization interval becomes

$$\Delta V = (2V_{max})/2^N$$

Performance Derivation

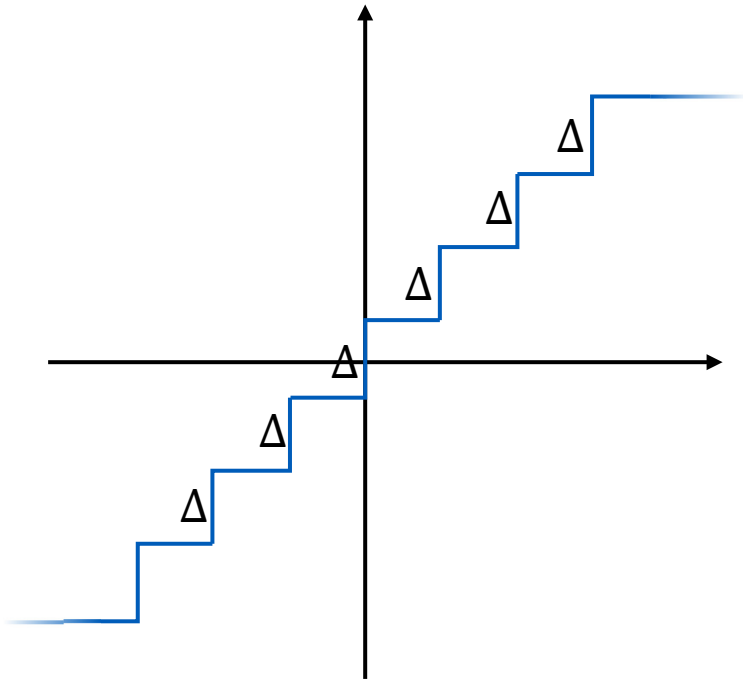
- Assume quantization accuracy of N bits per sample
- $SQNR$ becomes (worst case)

$$SQNR = 20\ log_{10}\frac{V_S}{V_{QN}} = 20\ log_{10}\frac{2^{N-1}\Delta}{\frac{\Delta}{2}} = 20\ log_{10}\frac{2^{N-1}}{\frac{1}{2}}$$

$$= 20\ log_{10}2 \cdot 2^{N-1} = 20\ log_{10}2^1 \cdot 2^{N-1}$$

$$= 20\ log_{10}2^N = 20\ N\ log_{10}2 \approx 6{,}02N\ dB$$

  - Maximum signal mapped to $2^{N-1} - 1\ (\simeq 2^{N-1})$
  - Most negative signal mapped to -2^{N-1},
  - This gives the **Peak SQNR**



SQNR properties

Given *SQNR* ≈ *6N* every extra bit gives

- twice the number of amplitude levels
  - $2^N \rightarrow 2^{N+1} = 2 \cdot 2^N$
- half of the quantization interval

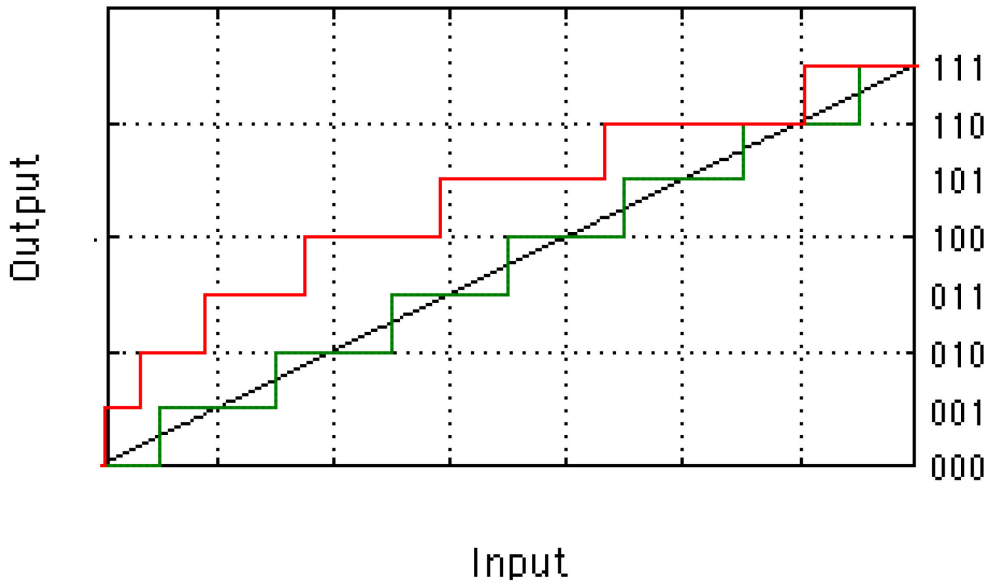
- a quarter of the quantization distortion power

Quantization interval spacing

Uniform quantization intervals

Non-uniform quantization intervals

- Smaller intervalls at certain ranges compared to others



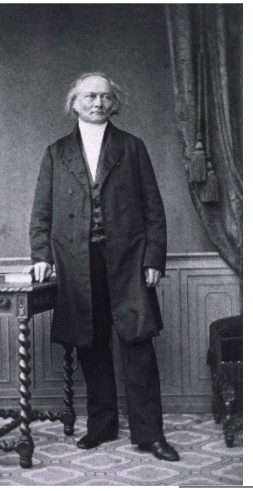
Why non-uniform spacing?

- More adopted to human senses
- Weber law

$$dp = k\frac{dS}{S} = k\frac{1}{S}ds$$

where

dp = differential change in perception
dS = differential increase in stimulus
S = starting value of the stimulus
k = constant proportionality



- Solving for p, using integration, we get
  $p = k \cdot ln(S) + C$

  At $p = 0$ we state that
  $C = -k \cdot ln(S_0)$
  where,
  S_0 then is a threshold below which no perception occurs.

- This gives

$$p = k \cdot ln\left(\frac{S}{S_0}\right)$$

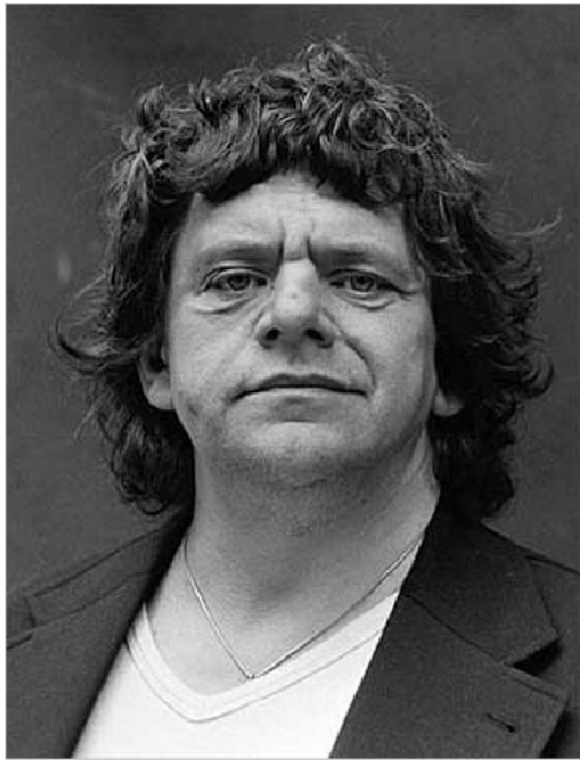
Relationship between stimulus and perception is logarithmic

  - Both perception of brightness (images) and sound intensity (audio) is logarithmic

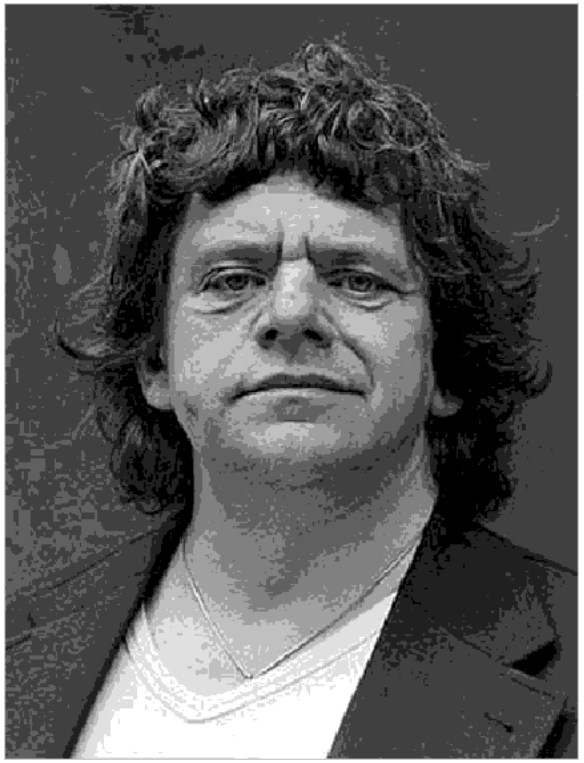
Audio quality vs bitrate

Quality	Sample Rate (Khz)	Bits per Sample	Channels	Data Rate (kB/sec)	Frequency Band (KHz)
Telephone	8	8	Mono	8	0.200-3.4
AM Radio	11.025	8	Mono	11.0	0.1-5.5
FM Radio	22.05	16	Stereo	88.2	0.02-11
5G audio	<48	16	Mono	<16	0.02-20
CD	44.1	16	Stereo	176.4	0.005-20
DVD Audio	192 (max)	24(max)	6 channels	1,200 (max)	0-96 (max)

Image quality vs bitrate (bits per pixel – bpp)



$8\ bpp$
$256\ gray\ levels$



$4\ bpp$
$16\ gray\ levels$



$2\ bpp$
$4\ gray\ levels$



$1\ bpp$
$2\ gray\ levels$

Demo

Distortion measures

In order to measure the amount of distortion three measures are commonly used

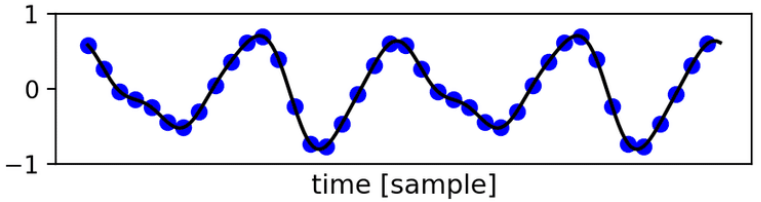
- Mean Squared Error (MSE) $\sigma_d^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - y_n)^2$
- Signal to Noise Error (SNR) $SNR = 10\log_{10}\frac{\sigma_x^2}{\sigma_d^2}$
- Peak SNR (PSNR) $PSNR = 10\log_{10}\frac{x_{peak}^2}{\sigma_d^2}$

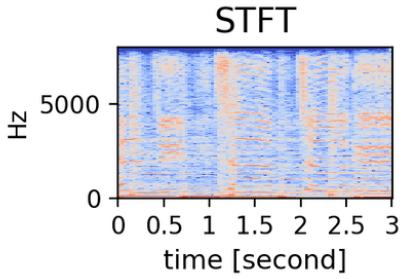
Begränsad delning

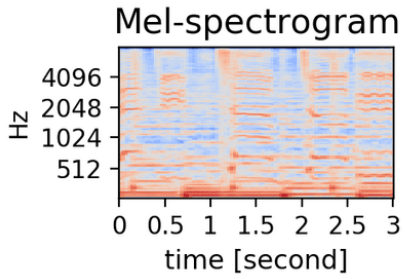
Representation

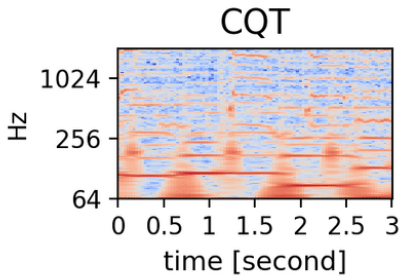
Signal representation scheme

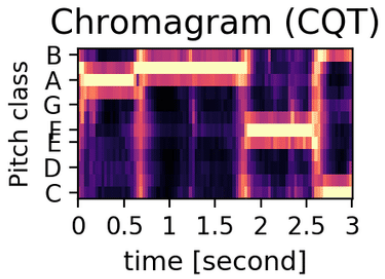
- Different representation formats serve different purposes











Audio examples

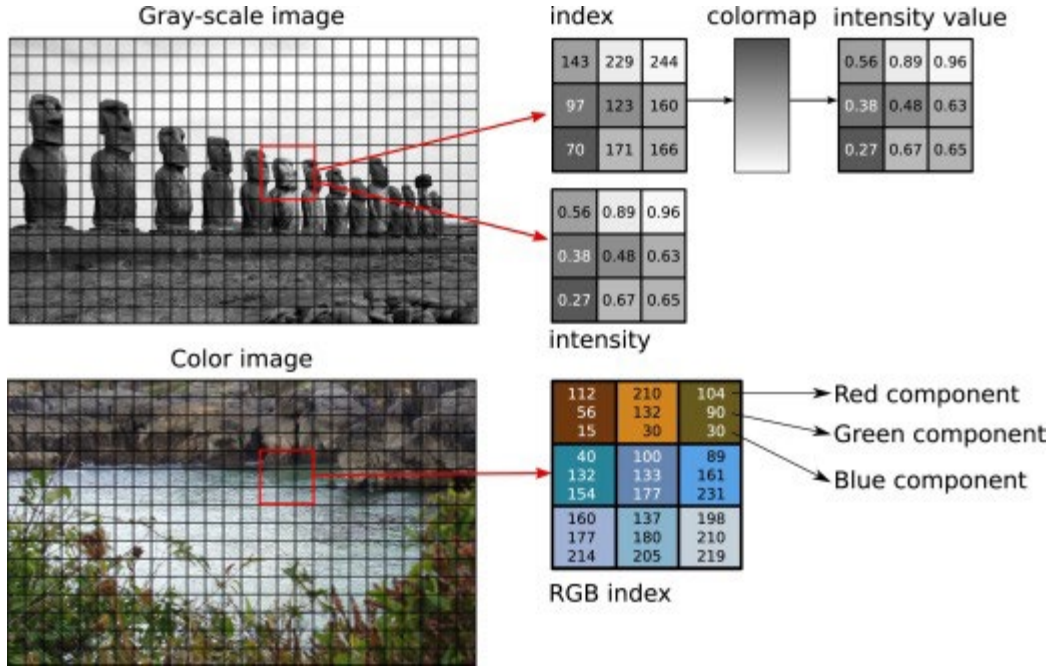
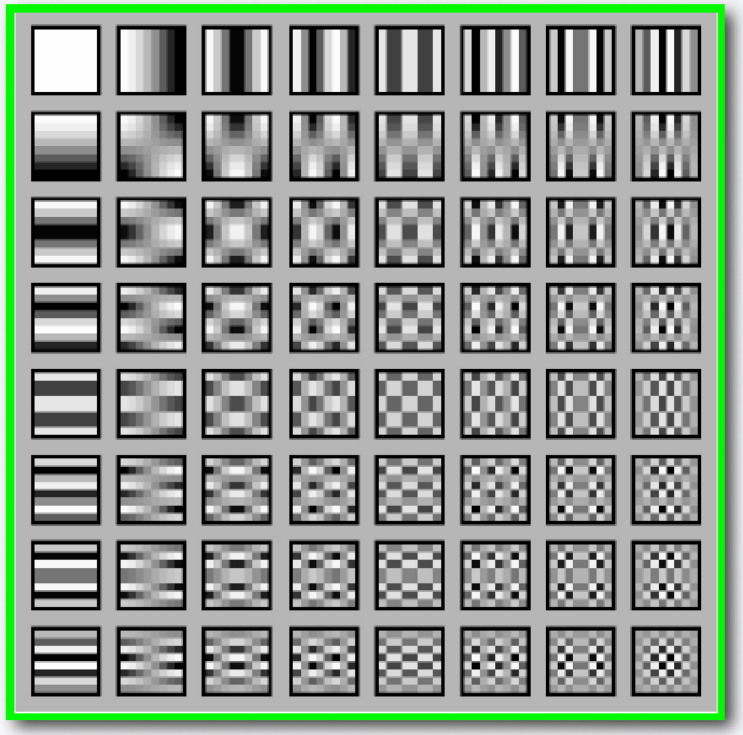


Image examples

Discrete Cosine Transform

$$F(u,v) = \frac{2\,C(u)\,C(v)}{\sqrt{MN}} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \cos\frac{(2i+1)\cdot u\pi}{2M} \cdot \cos\frac{(2j+1)\cdot v\pi}{2N} f(i,j)$$

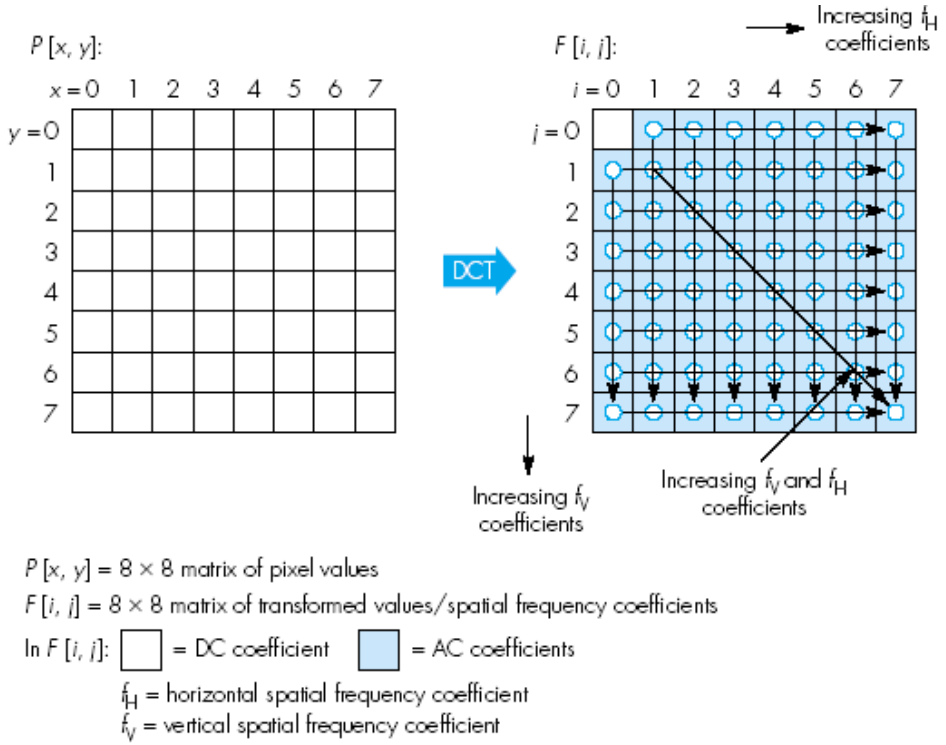
672	175	9	13	5	0	-1	-3
-281	-176	18	-5	-4	1	-1	1
150	28	-55	-5	-9	0	-2	0
-51	39	46	0	8	1	0	-5
1	-35	-15	15	1	0	0	-3
-7	13	-4	-7	1	-3	0	-3
7	-2	3	4	1	4	-3	-4
-5	2	0	7	0	-4	1	-2





Blockwise DCT

- Decompose the original signal into spatial frequencies, including DC and AC components

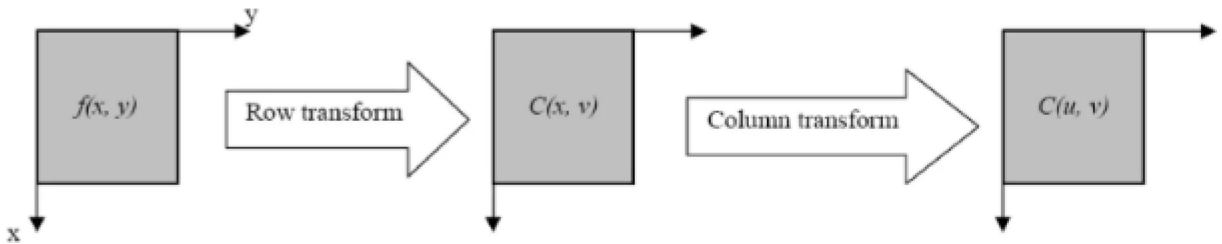


DCT – a separable base (*)

- The 2D 8x8 DCT can be separated into a sequence of two, 1D DCT steps:

$$G(i,v) = \frac{1}{2} C(v) \sum_{j=0}^{7} \cos \frac{(2j+1)v\pi}{16} f(i,j)$$

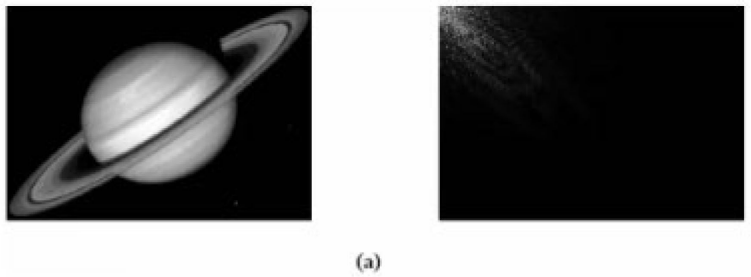
$$F(u,v) = \frac{1}{2} C(u) \sum_{i=0}^{7} \cos \frac{(2i+1)u\pi}{16} G(i,v)$$



- A separable transformation can be performed much faster, less computations are required

(*) as is the DFT

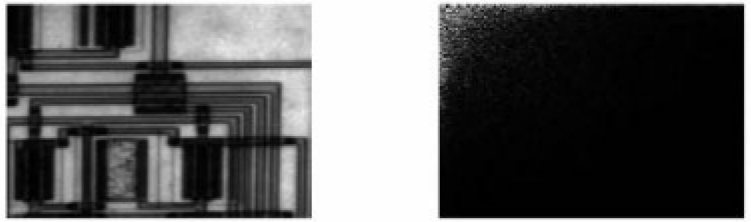
Energy compaction by using DCT

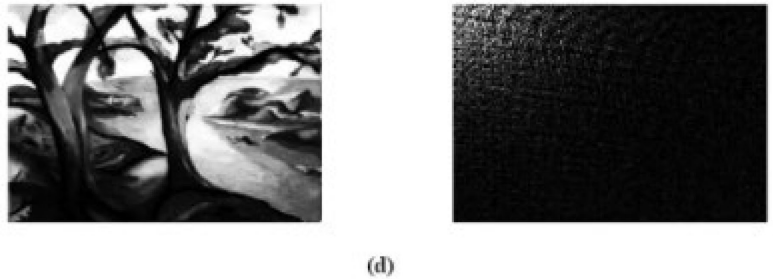


(a)

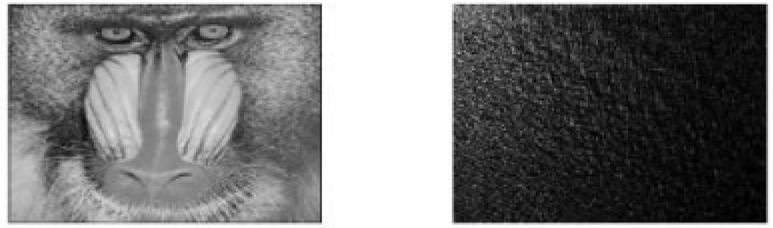


(b)

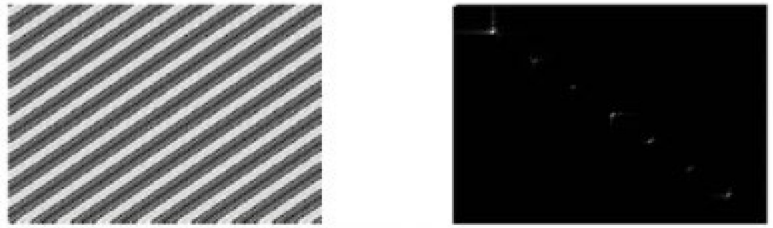




(d)

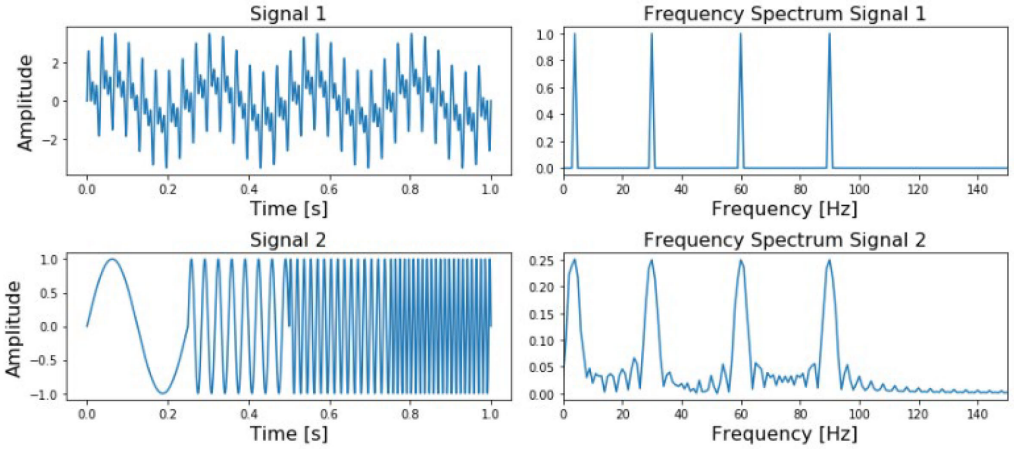


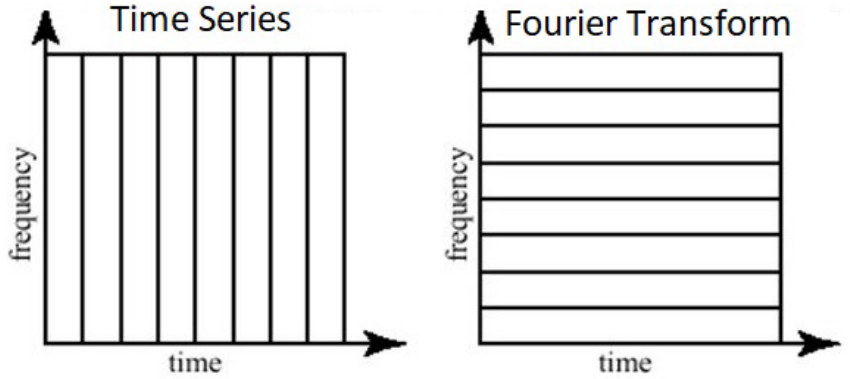
(e)



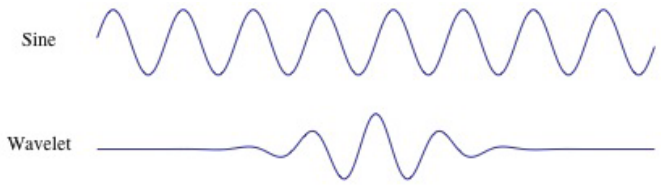
Wavelet

- Fourier transformation is localized in frequency only
  - No information about from what part of x(n) a specific frequency derives





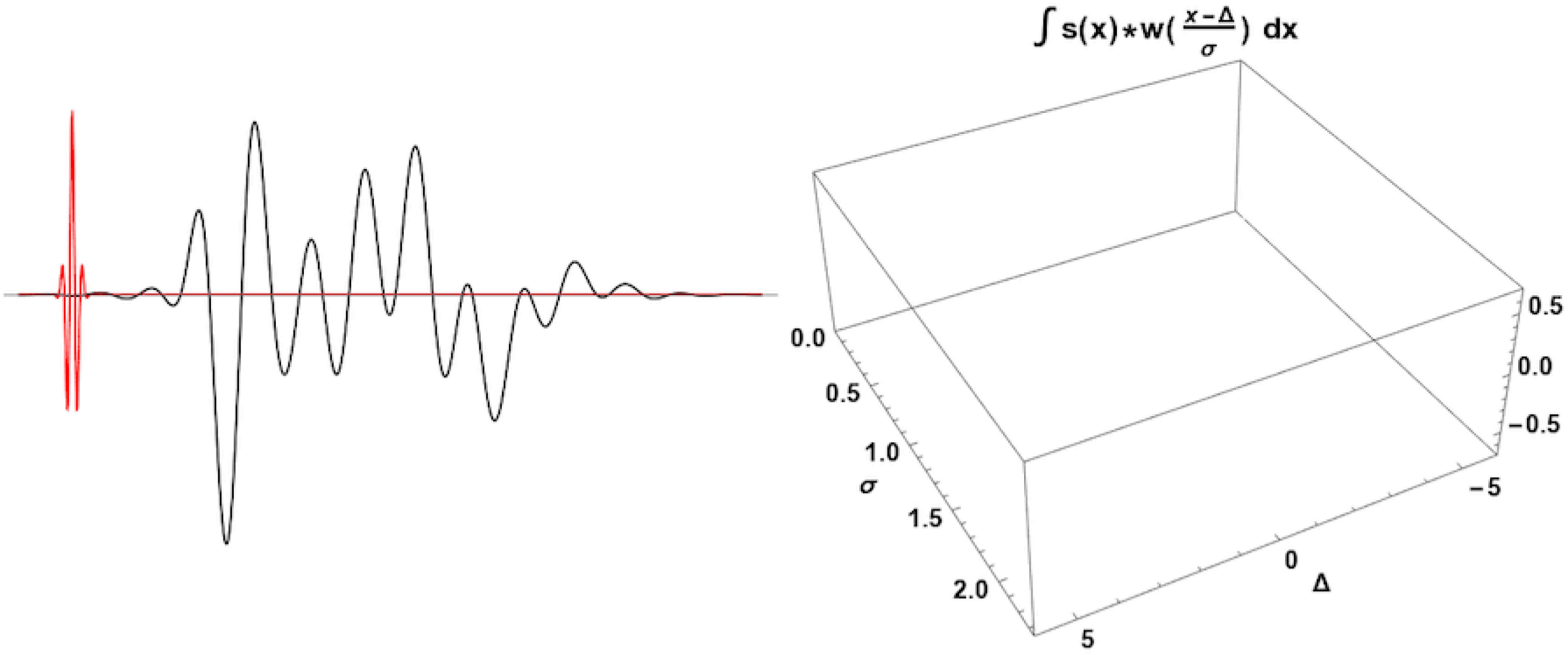
- Wavelet functions are localized in both time/space and frequency



Frequency is a single parameter

Wavelet cont.

- Scale σ and shift Δ are the two parameters
  (compare to Fourier Transform that has a single parameters – frequency)



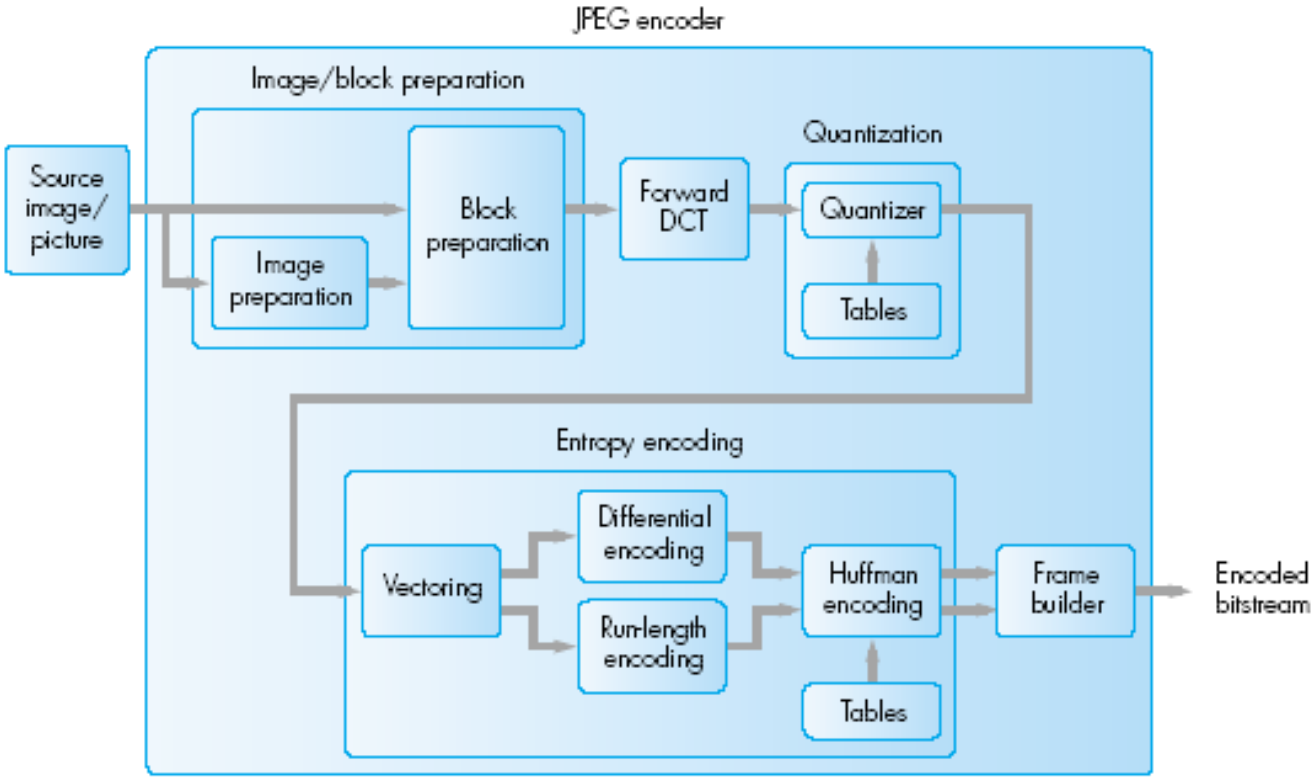
Begränsad delning

Formats

5. Data Formats in Signal and Image Processing (10 minutes)

- Overview of common file formats
  - Audio: WAV, MP3, AAC
  - Image: JPEG, PNG, TIFF
  - Video: MP4, AVI, MOV
  - Light field: ILLF, LFP

JPEG = Joint Photographic Experts Group



Block-wise quantization of DCT coefficients

- Scaled quantization

DCT coefficients

120	60	40	30	4	3	0	0
70	48	32	3	4	1	0	0
50	36	4	4	2	0	0	0
40	4	5	1	1	0	0	0
5	4	0	0	0	0	0	0
3	2	0	0	0	0	0	0
1	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Quantizer

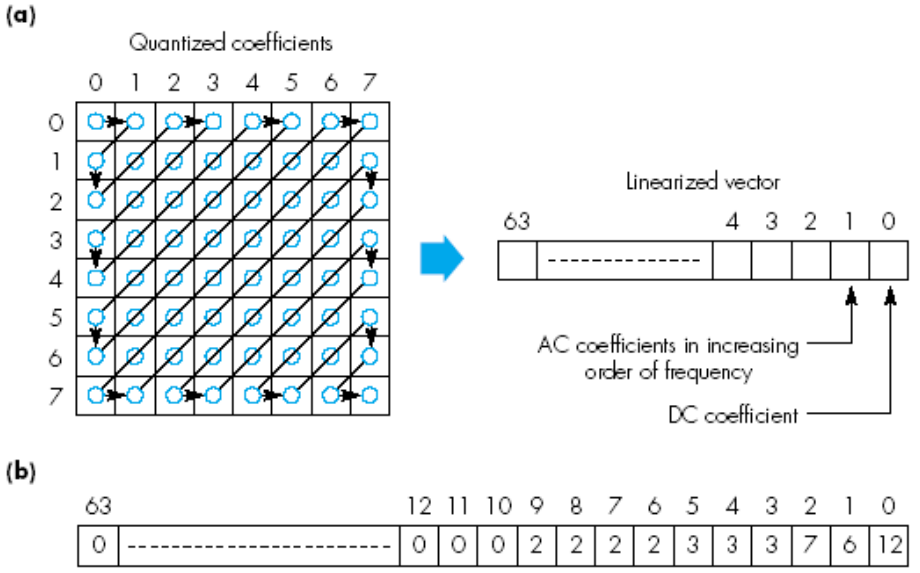
Quantized coefficients

12	6	3	2	0	0	0	0
7	3	2	0	0	0	0	0
3	2	0	0	0	0	0	0
2	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

10	10	15	20	25	30	35	40
10	15	20	25	30	35	40	50
15	.20	25	30	35	40	50	60
20	25	30	35	40	50	60	70
25	30	35	40	50	60	70	80
30	35	40	50	60	70	80	90
35	40	50	60	70	80	90	100
40	50	60	70	80	90	100	110

Quantization table

- Vectorization for later entropy encoding



Basic structures of formats

- Metadata
- Headers
- Data
- (Error correction - for use with lossy channels)